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1 Dealscan Variables

Feature	Why It Matters
loantype	Captures tranche structure (Revolver, Term Loan A/B, Bridge). Strong determinant of pricing.
primarypurpose	Indicates why the borrower is raising funds (acquisitions, refinancing, etc.). Risk varies by purpose.
secured	Collateral lowers perceived risk, leading to lower spreads.
seniority	Priority in repayment; higher seniority implies safer lending and lower spreads.
distributionmethod	Distinguishes “underwritten” vs. “best efforts” syndication; affects underwriting risk and pricing.
countryofsyndication	Jurisdictional and legal differences influence interest spreads.
dealstatus	Useful for filtering (e.g., completed vs. cancelled). Only completed deals are selected.
dealpurpose	Package-level complement to primarypurpose; often provides broader intent.
currency	Captures market-specific pricing norms (USD, EUR, GBP).
projectfinancetype	Specialized structure (e.g., infrastructure projects) with distinct pricing behavior.
facilityamt	Facility size — larger loans often have lower spreads due to borrower strength and economies of scale.
maturity	Longer maturities imply higher spreads (term premium).
dealamount	Package size — proxy for borrower size and negotiating power.
salesatclose	Borrower’s annual revenue at close — strong indicator of credit quality and spread level.
averagelife	Weighted average loan life — longer average implies greater risk.
baserate	Reference rate type (LIBOR, Prime, SOFR) — controls for market-rate context.
fee	Captures direct cost components affecting spread.
minbps / maxbps	Spread boundaries directly tied to all-in spread formation.
prorataallocation	Share of pro rata vs. institutional funding — affects investor mix and risk premium.

Table 1: Summary of Dealscan variables and their economic relevance.

1.1 Exploring Variation in All-in-Drawn Spreads Across Loan Features

One of the central objectives of the analysis is to understand how the all-in-drawn spread varies across major deal characteristics. Loan type is particularly informative because it embeds maturity, purpose, and market segment differences. Figure 1 illustrates the mean all-in-drawn spread for each loan type category in the sample.

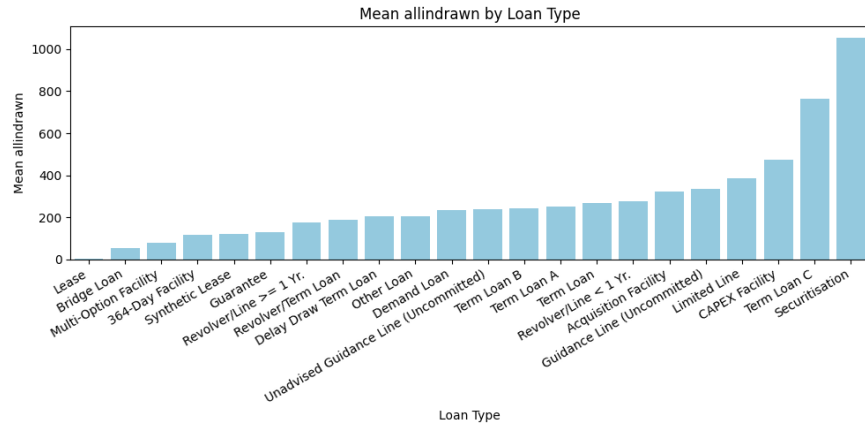


Figure 1: Mean all-in-drawn spread by loan type.

Interpretation

The figure highlights substantial heterogeneity in pricing across instruments. Short-term and liquidity-focused products—such as leases, bridge loans, 364-day facilities, multi-option lines, and synthetic leases—show the lowest all-in-drawn spreads, typically under 200 bps. These facilities are designed for working-capital flexibility rather than long-term leverage, resulting in lower perceived credit and duration risk.

As the loan structures become more leveraged or institutional in nature, spreads increase materially. Term Loan A and standard revolvers fall near the middle of the distribution, while Term Loan B and Term Loan C—core components of leveraged buyouts—exhibit significantly higher spreads aligned with their higher borrower risk profiles.

The upper end of the distribution includes CAPEX facilities, securitizations, and certain long-tenor institutional term loans, with average spreads reaching 800–1,000+ bps. These products typically involve higher leverage, weaker collateral structures, or borrowers operating in more stressed conditions.

Another important structural determinant of loan pricing is the position of the facility within the borrower’s capital structure. Seniority governs the order of repayment in distress, and therefore directly influences lender risk and required compensation. Figure 2 summarizes the mean all-in-drawn spread for each seniority tier in the Dealscan sample.

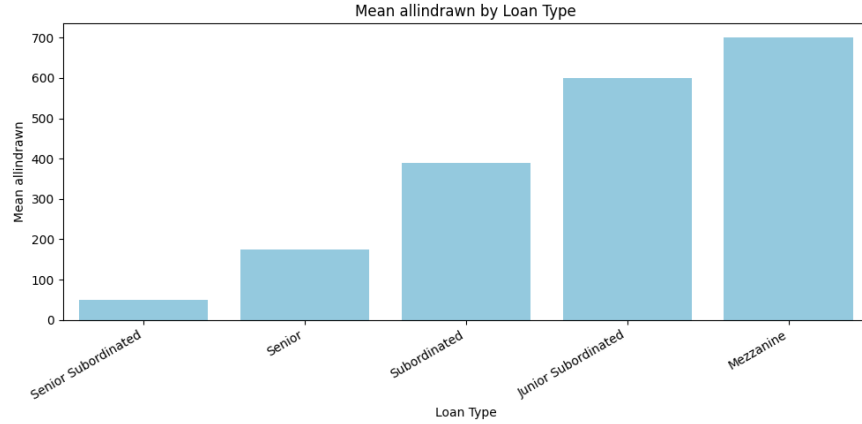


Figure 2: Mean all-in-drawn spread by seniority class.

Interpretation

The pricing gradient across seniority levels is clear and monotonic. Senior facilities exhibit the lowest all-in-drawn spreads, consistent with their priority of repayment and stronger collateral protection. Moving one level down the capital structure to subordinated debt produces a substantial increase in spreads, reflecting the greater loss severity faced by lenders in default scenarios.

Junior subordinated tranches and mezzanine debt sit at the bottom of the repayment waterfall and accordingly command the highest required returns. Mezzanine spreads in particular exceed 700 bps on average, aligning with their hybrid debt–equity risk profile and frequent use in highly leveraged transactions.

Taken together, Figure 2 reinforces a foundational empirical regularity: **lower repayment priority leads to sharply higher spreads**. This pattern mirrors both theoretical predictions from structural credit models and long-established evidence from corporate bond markets.

2 Macro Indicators

Indicator	Description
FEDFUNDS	Effective federal funds rate.
M2REAL	Inflation-adjusted M2 money supply.
T10Y2YM	10-year minus 2-year Treasury yield spread (term spread).
UNRATE	Unemployment rate.
USREC	US recession indicator (binary).
VIXCLS_last	Latest VIX index value (end of month).
credit_risk	Market credit risk measure (BAA–AAA credit spread).

Table 2: Macroeconomic indicators used in the analysis.

3 Compustat Variables

All Compustat accounting variables used in the analysis are log-scaled prior to modeling.

Variable	Description and Default-Risk Relevance
sale	Total revenue generated from goods or services sold during the period. Declining or volatile sales weaken operating stability and increase default risk.
oiadp	Operating income after depreciation. Strong operating profit indicates healthy cash-generating capacity, reducing the likelihood of default.
xint	Total interest expense. High interest burdens relative to earnings elevate financial strain and default probability.
ni	Net income after all expenses and taxes. Persistent losses erode equity buffers and increase default risk.
xsga	Selling, general, and administrative expenses. Excessive overhead reduces profitability and liquidity, raising default risk.
at	Total assets. Larger asset bases provide collateral and income-generating capacity, improving default resilience.
lt	Total liabilities. Higher liabilities mean greater repayment obligations and higher probability of distress.
ceq	Common equity. Equity serves as a loss-absorbing buffer; low equity magnifies leverage and default risk.
act	Current assets. Sufficient short-term assets improve liquidity and reduce vulnerability to default.
lct	Current liabilities. High short-term obligations relative to liquidity increase near-term default risk.
oanfc	Operating cash flow. Positive recurring cash flow strengthens financial flexibility and lowers default probability.
capx	Capital expenditures. Large investment spending can strain cash reserves and increase reliance on debt.
finfc	Financing cash flow. Heavy dependence on debt or equity issuance signals potential liquidity vulnerability.
ivncf	Investing cash flow. Persistent outflows reduce liquidity if not offset by strong operations.
chech	Cash and cash equivalents. Higher cash buffers improve solvency and reduce default risk.
cash_dividends_net_assets	Cash dividends relative to assets. Large payouts reduce retained earnings and liquidity, raising financial fragility.
investment_net_assets	Investment spending relative to assets. High investment intensity increases funding needs and possible distress if returns lag.
internal_cash_flow_net_assets	Internal operating cash flow scaled by assets. Strong internal funding lowers dependence on external financing and reduces default likelihood.
debt_book_cap	Total debt relative to book capitalization. High leverage increases fixed obligations and default risk.

Table 3: Compustat accounting variables (log-scaled) and their relevance for default and credit risk.

4 Beta Variables

This set of market-based variables captures equity return sensitivity, idiosyncratic volatility, and return-based performance measures. All variables are standardized prior to modeling.

Variable	Description and Credit-Risk Relevance
PERMNO	Unique CRSP firm identifier used to merge market data with accounting and loan information.
year	Observation year associated with the computed betas and volatility metrics.
industry	Industry classification used to control for sector-level differences in volatility and risk exposure.
beta12	Equity beta estimated using 12 months of returns. Higher betas imply stronger sensitivity to market conditions and greater cyclical risk.
beta24	Equity beta estimated using 24 months of returns. Provides a more stable long-run estimate of systematic risk.
beta36	Equity beta estimated using 36 months of returns. Longer windows capture persistent exposure to broad market movements.
tvol12	Total return volatility over 12 months. Higher total volatility increases uncertainty and perceived credit risk.
tvol24	Total return volatility over 24 months. Smoother measure of long-run volatility conditions.
tvol36	Total return volatility over 36 months. Captures sustained risk patterns important for lender pricing.
mvol12	Market (systematic) volatility component over 12 months. Reflects sensitivity to economy-wide shocks.
mvol24	Market volatility over 24 months. Provides a more stable assessment of systematic risk.
mvol36	Market volatility over 36 months. Useful for long-term credit risk profiling.
svol12	Idiosyncratic (firm-specific) volatility over 12 months. High idiosyncratic volatility signals uncertainty about firm fundamentals and higher credit risk.
svol24	Idiosyncratic volatility over 24 months. Smooths short-term noise while preserving firm-specific risk information.
svol36	Idiosyncratic volatility over 36 months. Persistent firm-specific uncertainty increases distress likelihood.
ivol12	Alternative idiosyncratic volatility measure computed using 12-month residual return variance.
ivol24	Idiosyncratic volatility using 24 months of data. Useful for detecting medium-term firm-specific risk.
ivol36	Idiosyncratic volatility using 36 months of data. Captures long-run idiosyncratic instability relevant for credit pricing.
returns12	Cumulative stock return over the prior 12 months. Poor performance increases refinancing and default risk.
returns24	Cumulative return over 24 months. Captures medium-term market sentiment and transition into financial distress.
returns36	Cumulative return over 36 months. Long-term underperformance often precedes deterioration in credit quality.

Table 4: Market-based beta and volatility variables used in the analysis.

4.1 Correlation Structure Across All Financial, Market, and Macro Variables

To obtain a comprehensive view of the relationships in the dataset, we compute a full correlation matrix that includes Dealscan quantitative variables, Compustat accounting variables (log-scaled), beta and volatility measures, and all macroeconomic indicators. This consolidated view helps identify clusters of related features, potential multicollinearity, and broader economic linkages relevant for credit pricing.

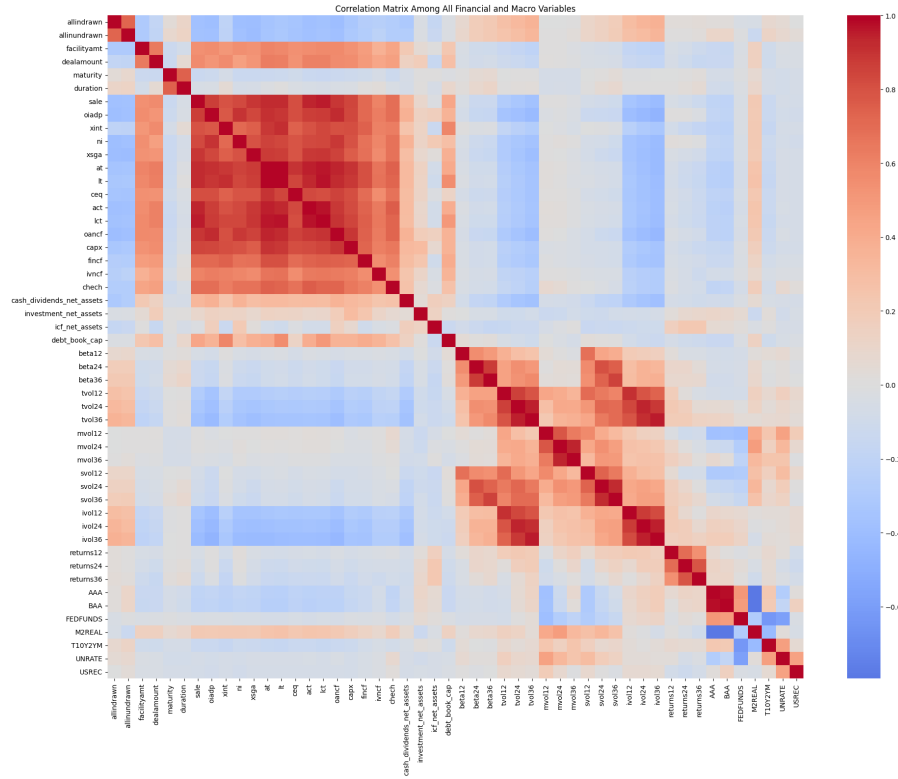


Figure 3: Correlation matrix among Dealscan quantitative variables, log-scaled Compustat fundamentals, beta/volatility measures, and macroeconomic indicators.

Interpretation

The heatmap in Figure 3 shows several clear correlation structures:

- **Fundamental block.** Compustat variables cluster tightly, reflecting firm size and accounting linkages.
- **Beta/volatility block.** Beta and volatility measures form consistent horizon-based clusters, capturing systematic and idiosyncratic risk.
- **Return patterns.** Lagged returns are negatively correlated with volatility and leverage, consistent with distressed-firm dynamics.
- **Loan structure.** Deal amount, facility size, maturity, and duration correlate moderately among themselves but weakly with fundamentals.
- **Macro indicators.** AAA/BAA spreads, yield curve slope, FEDFUNDS, and recession flags exhibit modest but coherent correlations with market risk variables.

- **All-in-drawn spreads.** The dependent variable is positively related to risk measures (volatility, leverage) and negatively related to size fundamentals.

Overall, the matrix reveals distinct clusters—fundamentals, market risk, loan characteristics, and macro conditions—indicating meaningful structure and manageable multicollinearity across features.

5 Results

Model	Train RMSE	Test RMSE	Train R^2	Test R^2
Lasso Regression	55.2305	58.8013	0.4955	0.0566
XGBoost Regression	18.7338	40.0140	0.9420	0.5631
KNN Regression	2.5320	44.2680	0.9989	0.4653
LightGBM Regression	24.3753	38.7115	0.9017	0.5911
PyTorch MLP Regression	36.3205	51.9798	0.7818	0.2628
Tuned MLP (Hidden=64, LR=0.0005, ReLU)	0.6008	0.7368	0.7431	0.2973
Allindrawn (main task)	0.6069	0.6472	0.6316	0.3091
Allinundrawn (auxiliary)	0.5725	0.5624	0.6722	0.2023

Table 5: Model performance summary across regression approaches, including hyperparameter-tuned MLP.

Effect of the Auxiliary Task

Main-task performance. Including the auxiliary target (allinundrawn) led to a modest improvement relative to the single-task MLP. The single-task MLP achieved a test RMSE of 0.7368 and $R^2 = 0.2973$, whereas the multi-task setting slightly improved the error to 0.6472 with $R^2 = 0.3091$. Thus, multi-task learning reduced prediction error and improved generalization.

Loss behavior. The allindrawn loss decreased more smoothly during training in the multi-task model, with less variance across epochs. This suggests that the auxiliary signal provided an additional source of regularization, stabilizing the learned representation.

Feature importance and representation. The multi-task model gives more weight to features tied to borrower fundamentals and loan structure, aligning with variables important for both drawn and undrawn spread determination. This indicates that the shared layers learned a representation capturing common economic drivers of both outcomes (e.g., maturity, facility amount, risk indicators).

Interpretation. The performance improvement likely arises because allindrawn and allinundrawn share underlying credit-risk determinants. Joint training forces the network to learn features that generalize across related tasks, reducing overfitting and acting as an inductive bias. This is consistent with multi-task learning theory: when tasks are correlated, sharing parameters improves sample efficiency and stabilizes learning.

Conclusion. Adding the auxiliary task improved the main task’s predictive accuracy, stabilized training, and encouraged more economically coherent

feature use. The empirical evidence suggests that allinundrawn provides meaningful complementary information for estimating allindrawn, making multi-task learning a beneficial approach in this setting.

6 Hyperparameter Search

A grid search over hidden dimensions (64, 128, 256), learning rates (0.001, 0.0005, 0.0001), and activation functions (ReLU, Tanh, Sigmoid, Linear) was conducted. Each configuration was trained for 100 epochs.

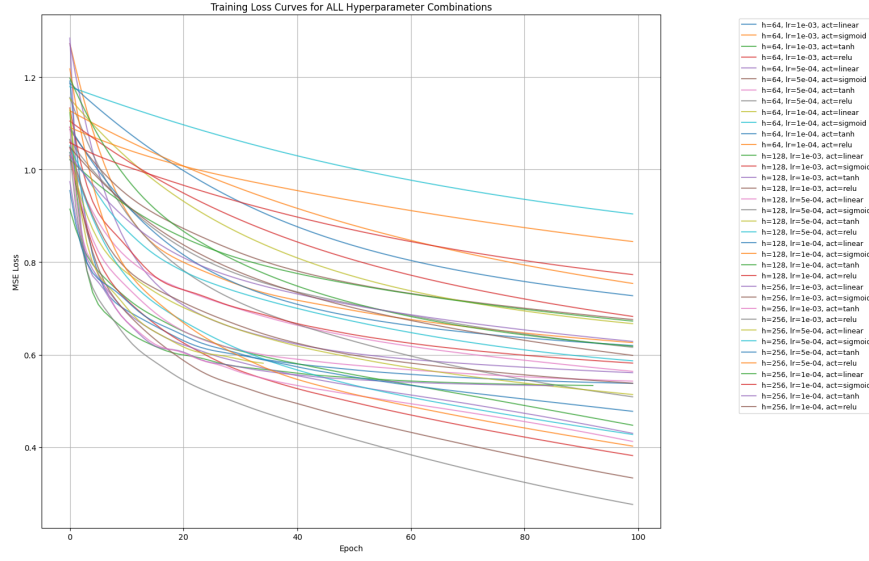


Figure 4: Training loss curves for all hyperparameter combinations.

Summary

Most configurations converge smoothly, with lower learning rates producing more stable curves. ReLU generally outperforms other activations, and increasing hidden size provides gradual but limited gains.

Top 5 Configurations by Test RMSE

Hidden Dim	LR	Act	Test RMSE	Val Loss
64	0.0005	ReLU	0.6526	0.2939
256	0.0005	Linear	0.6536	0.3160
128	0.0001	Tanh	0.6573	0.3848
256	0.0001	Linear	0.6636	0.3481
64	0.0001	Tanh	0.6660	0.4063

Table 6: Top 5 hyperparameter configurations by lowest test RMSE.

Top 5 Configurations by Test R^2

Hidden Dim	LR	Act	Test R^2	Val Loss
64	0.0005	ReLU	0.2973	0.2939
256	0.0005	Linear	0.2953	0.3160
128	0.0001	Tanh	0.2873	0.3848
256	0.0001	Linear	0.2735	0.3481
64	0.0001	Tanh	0.2682	0.4063

Table 7: Top 5 hyperparameter configurations by highest test R^2 .

Best Overall

The best-performing model is:

$$Hiddendim = 64, \quad LR = 0.0005, \quad Activation = ReLU.$$